Foundations of Artificial Intelligence - 2026.1

Atomic Decisions under Risk

Felix Mohr

Universidad de La Sabana

Chía, March 3, 2026

Class Overview

1. Atomic Environments under Risk or Uncertainty

2. Making Rational Decisions in Atomic Environments under Risk

3. The Bernoulli Principle

Course Outline

1. Atomic Environments under Risk or Uncertainty
2. Making Rational Decisions in Atomic Environments under Risk
3. The Bernoulli Principle

One-Shot Decision Problems

We now consider an environment of the following type:

- a state space $S = \{s_0\} \cup S'$ consisting of
  - an initial state s_0, and
  - possible final states S' after a decision is made,
- a set of events E that is determined by a set A of possible actions (decisions)
- the transition ψ is non-deterministic and is
  - only defined for s_0 and the actions $a \in A$ and
  - governed by a conditional probability distribution $\mathbb{P}(S'|a)$, one for each $a \in A$

One-Short Decision Problems under Risk or Uncertainty

If the agent knows $\mathbb{P}$, then we have a decision under risk, otherwise under uncertainty.

We need to clarify two points:

1. What is the problem to be resolved?
2. To understand that, we need to really understand what the notation $\mathbb{P}(S' \mid a)$ means mathematically.

Discrete Probability Measure

Let Ω be some countable set. We call the elements in Ω elementary events.

Then a probability mass distribution over Ω is a mapping

$$f : \Omega \longrightarrow [0, 1],$$

such that $f(\omega) \geq 0$ and $\sum_{\omega \in \Omega} f(\omega) = 1$.

Stochastic Events

Any subset $A \subseteq \Omega$ is a (stochastic) event. Elementary events are a special case: $|A| = 1$.

When we draw a random $\omega \in \Omega$ (according to f), we say that A occurred if $\omega \in A$.

To express the probability of an event, we define a probability measure

$$\mathbb{P} : 2^\Omega \to [0, 1] \quad \text{with} \quad \mathbb{P}(A) := \sum_{\omega_i \in A} f(\omega_i).$$

In particular, we get that $(\Omega, \mathbb{P})$ form a discrete probability space, because

- $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$ and
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ for $A, B \subseteq \Omega$ with $A \cap B = \emptyset$.

Stochastic Events

Example

Example of rolling a dice:

- $\Omega = \{1, \ldots, 6\}$,
- Event "more than two eyes": $A = \{3, 4, 5, 6\}$
- Rolling the dice determines a random assignment for ω
- The value of ω define whether event A occurs:
  - $\omega = 5$ means that A did occur
  - $\omega = 2$ means that A did not occur
- $\mathbb{P}(A) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6}$

Conditional Probability

The conditional probability $\mathbb{P}(A \mid B)$ refers to the probability that event A happens given that B is known to happen:

$$\mathbb{P}(A \mid B) \stackrel{\text{df}}{=} \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

$\mathbb{P}(A \mid B)$ is only defined for $\mathbb{P}(B) > 0$.

For any B with $\mathbb{P}(B) > 0$, the tuple $(\Omega, \mathbb{P}(\cdot \mid B))$ is a discrete probability space again.

Independency

Two events A and B are intuitively independent if knowing the result of A does not change the probability of B and vice versa.

This can be formalized as follows:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \quad \text{and} \quad \mathbb{P}(B \mid A) = \mathbb{P}(B)$$

This is equivalent to the definition of stochastic independence of A and B:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

if $\mathbb{P}(A) > 0$ bzw. $\mathbb{P}(B) > 0$.

Random Variables

Let $(\Omega, \mathbb{P})$ be a discrete probability space.

A discrete random variable (RV) is any function

$$X : \Omega \longrightarrow \Omega',$$

where Ω' is the set of possible values of the RV X.

If $\Omega' \subseteq \mathbb{R}$, then X is called real-valued.

Even if $\Omega' = \mathbb{R}$, the RV X is still discrete since Ω only allows a discrete number of cases.

Random Variables
Example

Consider the following game:

- We roll a fair dice.
- You win 1\$, 2\$ or 3\$ if the dice shows 4, 5, or 6.
- Otherwise you don't win anything.

Formally,

- we have a discrete prob. space $\Omega = \{1, .., 6\}$, and $\mathbb{P}(\omega) = \frac{1}{6}$ for all $\omega \in \Omega$,
- a result space $\Omega' = \{0, 1, 2, 3\}$, and
- we can model the result as a RV $X(\omega) := \max\{0, \omega - 3\}$.

Random Variables
Assertions of Values as Events

For some RV X, any assertion about its values is true or false for some ω, e.g.:

- $X = 5$
- $X \leq 42$
- $X \in \{red, blue\}$

So $X = 5$ reads like a condition but actually *defines* a set of $\omega \in \Omega$ that satisfy it:

$$X = 5 \equiv \{\omega \in \Omega \,|\, X(\omega) = 5\}$$

Hence each such statement encodes an *event* in Ω.

Random Variables

The equality event $X = x$ for any $x \in \Omega'$ is important, because

$$\mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega \,|\, X(\omega) = x\}) = \sum_{\omega\,:\,X(\omega)=x} \mathbb{P}(\{\omega\}).$$

Because of this importance one also writes $\mathbb{P}(x) := \mathbb{P}(X = x)$.

And from this we also have

$$\mathbb{P}(X \in A) = \sum_{x \in A} \mathbb{P}(x) = \sum_{\omega\,:\,X(\omega)\in A} \mathbb{P}(\{\omega\}),$$

for any $A \subseteq \Omega'$, so $\mathbb{P}(X \in \cdot)$ is a probability measure in Ω'.

Random Variables

Examples for Probabilities

Example: Roulette with a bet of 120$ on red.

Modeled as:

- $\Omega = \{red, black, green\}$, $\mathbb{P}(red) = \mathbb{P}(black) = \frac{18}{37}$ and $\mathbb{P}(green) = \frac{1}{37}$
- $X(\omega) = \begin{cases} +120 & \text{if } \omega = red \\ -120 & \text{else} \end{cases}$

The new probability space is $\Omega' = \{0, 120\}$ and

- $\mathbb{P}(X = -120) = \mathbb{P}(\{black, green\}) = \frac{18}{37} + \frac{1}{37} = \frac{19}{37}$
- $\mathbb{P}(X = +120) = \mathbb{P}(\{red\}) = \frac{18}{37}$

Random Variables
Remarks on Random Variables

1. RVs are not random. What makes them assume random values is the random input.

   Given the random elementary input event ω, the value of the RV is *deterministic*.

2. RVs are nothing else than bridges between probability spaces.

   The variable defines the new elementary space Ω' and also induces the probability measure on it inherited from the original probability space $(\Omega, \mathbb{P})$.

   In the Roulette example, Ω can be seen as the image of a RV that maps from $\{0, .., 36\}$ to $\{\textit{red}, \textit{black}, \textit{green}\}$, and $\mathbb{P}$ induced from the uniform distribution.

Random Variables as Names for Distributions

In practice we often have $\Omega = D_1 \times .. \times D_d$, and we have d RVs with $X_i(\omega) = \omega_i$.

Example: In 3 rolls of a dice, where the 3 outcomes are X_1, X_2, X_3.

So RVs just project slices from Ω, and it is then easier to simply talk about the RVs, and Ω and $\mathbb{P}$, which is anyway never specified concretely, are left implicit.

In the example, $\Omega = \{1, .., 6\}^3$ is implicit, and $\mathbb{P}$ exists.

Conclusion:

- Theory: First the probability space and then the RV as a function.
- Practical: Define your RVs and together these imply a probability space.

Random Variables as Names for Distributions

The probability measure for all possible values of a single RV is called its distribution.

For a discrete RV X, the distribution is written as $\mathbb{P}(X)$ and contains the value $\mathbb{P}(x)$ for all possible values in D_X.

Mind the difference:

- $\mathbb{P}(x)$, which is $\mathbb{P}(X = x)$ is a *number*, but
- $\mathbb{P}(X)$, i.e., without a concrete value is a *collection*.

For finite variables, we can store the distribution $\mathbb{P}(X)$ in a vector or dictionary (where values of D_X are keys, and the value of x is $\mathbb{P}(x)$).

Expected Value

The expected value of a RV X with values in D_X is

$$\mathbb{E}[X] := \sum_{x \in D_X} x \cdot \mathbb{P}(x)\,.$$

For the conditional expected value of X on some event c, we can also write:

$$\mathbb{E}[X|c] := \sum_{x \in \mathbb{D}_X} x \cdot \mathbb{P}(x|c)\,.$$

One-Short Decision Problems under Risk or Uncertainty

In our decision making setup, we said that

> ... ψ ... *is governed by a conditional probability distribution* $\mathbb{P}(S'|a)$

If we consider two RVs, namely S' and A, then $\mathbb{P}(S' \mid a)$ is clearly defined.

Hereby we overwrite a bit the meaning of S' and A:

1. initially: the *set* of possible (i) successor states and (ii) actions
2. now: *RVs* that describe what happens (or what we do)

Course Outline

Rational Atomic Decisions under Risk

The agent wants to end up in the best possible next state s' (notion of $\succeq$).

Problem: The agent cannot pick a state but just an action.

Suppose you have 200$ and may invest all, half, or none of it into some project, which may succeed or fail with equal probability. The bank account state would be:

	p_1	p_2
a_1	200	200
a_2	100	300
a_3	0	500

Which action should be chosen?

Rational Atomic Decisions under Risk
Expected Outcome - Example

How can we transfer this situation into a formal model?

Here, S' is the outcome: $s_1 = 200, s_2 = 100, s_3 = 300, s_4 = 0, s_5 = 500$.

Thinking of the action $A \in \{a_1, a_2, a_3\}$ as a variable as well, we obtain

- $\mathbb{P}(s_1|a_1) = 1$, and
- $\mathbb{P}(s_2|a_2) = \mathbb{P}(s_3|a_2) = \mathbb{P}(s_4|a_3) = \mathbb{P}(s_5|a_3) = 0.5$

Clearly, $s_5 \succeq s_3 \succeq s_1 \succeq s_2 \succeq s_4$, but how does this translate to an ordering on A?

Rational Atomic Decisions under Risk

Expected Outcome

One idea is to look at expected outcome for an action a:

$$\mathbb{E}[S'|a] = \sum_{s' \in S} s' \cdot \mathbb{P}(s'|a)$$

So an action that maximizes the expected outcome is

$$a^* = \arg\max_{a \in S} \; \mathbb{E}[S'|a]$$

Rational Atomic Decisions under Risk

Expected Outcome - Example

Here, S' is the outcome: $s_1 = 200, s_2 = 100, s_3 = 300, s_4 = 0, s_5 = 500$.

Thinking of the action $A \in \{a_1, a_2, a_3\}$ as a variable as well, we obtain

- $\mathbb{P}(s_1|a_1) = 1$, and
- $\mathbb{P}(s_2|a_2) = \mathbb{P}(s_3|a_2) = \mathbb{P}(s_4|a_3) = \mathbb{P}(s_5|a_3) = 0.5$

So the action that maximizes the expected outcome is

$$a^* = \arg\max_{a \in A} \; \mathbb{E}[S'|a] = a_3$$

Course Outline

1. Atomic Environments under Risk or Uncertainty
2. Making Rational Decisions in Atomic Environments under Risk
3. The Bernoulli Principle

The Bernoulli Principle

The St. Petersburg Paradox

Let a coin be flipped until (in the n-th toss) we observe heads. The pay-off is then 2^n. The agent may pay E\$ to play this game or not play.

The expected outcome for playing is

$$\sum_{n=1}^{\infty} 2^n \cdot (1/2)^n - E = \infty \ ,$$

No matter how much we must invest or already possess, we can always expect the game to lift us to infinity.

Most players are only willing to invest small sums. Is there a flaw?

The Bernoulli Principle

The problem here is that having x times more is not x times better.

We capture the agent's preferences in a utility function $U : S \to \mathbb{R}$.

The utility function solves two problems at once:

1. A state that has x times higher utility *is* x times better.
2. The utility is always numeric and has an expected value, even if elements of the state space S are not numbers.

The Bernoulli Principle

For any action $a \in A$, we can now compute the expected utility

$$q(a) := \mathbb{E}[U(S')|s_0, a] = \sum_{s' \in S} U(s') \cdot \mathbb{P}(s'|s_0, a),$$

where q (quality) is just a notational shortcut emphasizing the functional character.

In this simple decision making situation, one often omits the state notation by

- omitting the (anyway constant) initial state s_0 and
- directly referring to the utility $U(S')$ as the (random) reward R, which yields

$$q(a) = \mathbb{E}[R|a] = \sum_{r \in domain(R)} r \cdot \mathbb{P}(r|a)$$

The Bernoulli Principle

The Bernoulli Principle says that the agent is rational iff he chooses an

$$a^* = \arg\max_{a \in A} \; q(a)$$

This can be seen positively or normatively.

A is the space $\mathcal{X}$ of decision objects and q induces an order over it.

Bernoulli Principle

Example

	p_1	p_2
a_1	200	200
a_2	100	300
a_3	0	500

Defining the utility function as $U(s) = \ln(s+1)$, then we obtain

$$q(a_1) \approx 5.3, \quad q(a_2) \approx 5.17, \quad q(a_3) \approx 3.11\ ,$$

such that action a_1 is optimal.